MC-EvoBench: Auditing Experience-Driven Self-Improvement in Minecraft Agents

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HarnessRSI 74 ★★★

evidence: task

Paper

Code

1 What is tested, what counts as a gain

Agent under test: JARVIS-1 online memory

Text planner + controller
plans 1-6 prompts; reflection writes one sentence of advice

JSON memory (the artifact)
plan, prompts, status, advice; latest 256 kept, top 4 used

Source phase: write memory, same env-step budget

Related
same pattern, other end item

Unrelated
collection tasks outside the chain

Shuffled
related tasks, reversed order

900 / 2,250 / 3,600 env steps per family (wood / wooden tool / stone tool)

RESET: world, inventory, working state
only the frozen artifact crosses (target memory read-only)

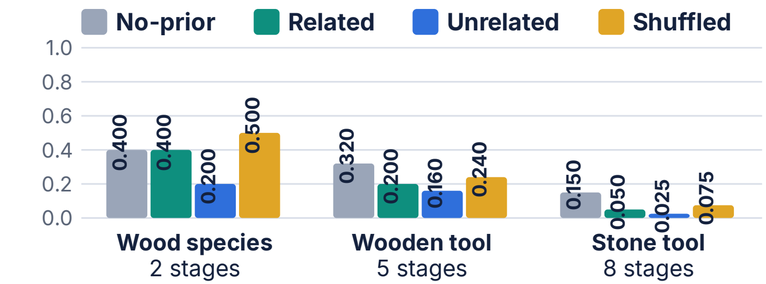
Target phase: same task, seed, verifier, step cap

No-prior | Related | Unrelated | Shuffled

Stage verifier: TSR · WGP · VMP, five seeds
gain counts only if related beats no-prior
AND the unrelated / shuffled controls
with paired 95% intervals

2 Results: no stable gain from related experience

Target TSR after reset, five seeds (Table 3)

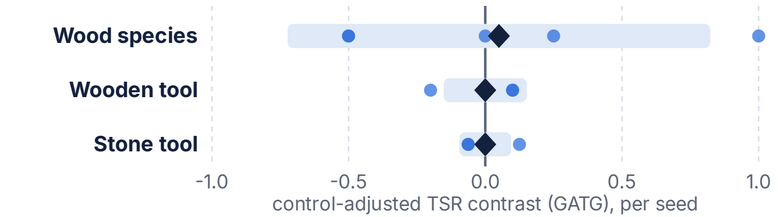


4 Where the gain is absent or unjudgeable

- Tool probes: **related point estimates below no-prior** on all 3 metrics (stone VMP 0.200 vs 0.533), within noise; wood: only WGP up (0.400 vs 0.360)
- Wood: **shuffled 0.500 ≥ related 0.400** (TSR); +0.050 GATG is one seed, **-0.19 without it**
- Stone-tool TSR is 0.150 even for no-prior (per seed 0.125 ×4, 0.250 ×1). Direct matrix (Fig. 3a): none of the reports reaching the crafting table passed it
- One method, 3 probes, 5 seeds, one world seed (31)
- Table 5: unrelated sources list passed=6 / 1 yet source TSR 0.000

3 Controls, cost and the execution floor

Per-seed TSR contrast (GATG), mean and 95% CI (Tables 4, D3)



Every GATG interval includes zero; wood species [-0.723, 0.823]. Largest GATG anywhere: 0.060. No CI is given for related minus no-prior.

Equal env steps, unequal computation (Table 5)

Source cost	source TSR (R)	planner calls R / U / S
Wood species	0.100	4.40 / 5.80 / 4.00
Wooden tool	0.080	14.80 / 12.00 / 10.00
Stone tool	0.150	**24.80 / 16.40 / 16.00**

More related-source calls on the two longer probes only; no target gain.

Direct capability, average TSR in % (Table 2)



Planner rows as in Table 2; paired probes use a planner-backed agent.

Takeaway

For this one agent, related experience showed no stable gain over no-prior or the matched controls after a world reset. The only positive contrast rests on one seed and all GATG intervals span zero. The test is small; other agents or budgets are not ruled out.

Our read · HarnessRSI

Score 74 · ★★★ · evidence level: task

Reports: reset world (same seed), 4 conditions, 5 seeds + CIs, cost.
Not reported: matched compute, other methods, leakage test.